P R O J E C T R E Q U I R E M E N T S
D O C U M E N T

AI Voice Agent Platform

Build, Train & Deploy Intelligent Voice Agents

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1. Executive Summary

This document outlines the complete project requirements for building an AI Voice Agent Platform — a self-service SaaS application that enables businesses and individuals to create, configure, train, and deploy intelligent voice-based AI agents powered by ElevenLabs Conversational AI, Claude API for natural language reasoning, and Supabase for data infrastructure.

The platform addresses the rapidly growing demand for voice AI solutions across industries including healthcare, customer support, sales, education, and hospitality. With the conversational AI market projected to reach $14.29 billion in 2025 and growing at 23.7% CAGR, this product is positioned to capture the emerging wave of businesses seeking accessible, no-code voice agent creation tools.

Users will be able to sign up, select a business domain, configure agent personality and voice characteristics, upload training data, and test their agent — all from a single dashboard. The platform will be deployable on any server infrastructure and support multi-tenant usage.

Key Value Propositions

- No-code voice agent creation with domain-specific intelligence
- ElevenLabs-powered natural voice synthesis with 5,000+ voice options
- Claude API-powered reasoning for context-aware, intelligent conversations
- Real-time agent testing with full conversation transcript visibility
- Multi-tenant architecture supporting unlimited users and organizations
- Self-hosted deployment model for complete data sovereignty

Suggested Product Names

Below are curated product name options across different branding strategies. Each name is evaluated for domain availability potential, memorability, and market positioning.

#	Product Name	Tagline	Positioning
1	VoxForge AI	Forge Your Voice Agent	Technical/builder-focused, implies crafting custom agents
2	AgentVox	Voice Agents, Your Way	Clean, professional, agent-first branding
3	VoiceNest	Hatch Intelligent Voice Agents	Warm, nurturing metaphor for building agents
4	SpeakCraft AI	Craft Conversations That Convert	Emphasizes customization and business outcomes

#	Product Name	Tagline	Positioning
5	TalkDeploy	Build. Train. Deploy. Talk.	Action-oriented, developer-friendly branding
6	VoxaHub	Your Voice Agent Command Center	Platform/hub positioning, central management
7	EchoMind AI	Intelligent Voice, Infinite Reach	Premium feel, AI intelligence emphasis
8	CallForge	Forge Smarter Conversations	Direct, business-focused, implies productivity

2. Market Research & Industry Trends

2.1 Market Overview

The AI voice agent market is experiencing unprecedented growth. The conversational AI market is projected to expand from $7.63 billion in 2025 to $139 billion by 2033 for voice AI agents specifically. Enterprise AI spending has reached $391 billion globally, with 92% of companies planning substantial generative AI investments over the next three years.

Metric	Value	Source Year
Conversational AI Market Size (2025)	$14.29 Billion	2025
Projected Market Size (2030)	$41.39 Billion	2030
CAGR Growth Rate	23.7%	2025–2030
Voice AI Agent Market (2033)	$139 Billion	2033
Enterprise AI Spending (Global)	$391 Billion	2025
Companies Planning GenAI Investment	92%	2025
Consumer Preference for AI Agents	44%	2025
YC Voice Agent Companies (Since 2020)	90+	2025

2.2 Key Industry Trends to Incorporate

2.2.1 Emotional Intelligence & Tone Adaptation

Modern voice agents detect emotions in speech and adjust their delivery accordingly. This reduces agent escalation by 25% and significantly improves customer satisfaction. Our platform should allow users to configure emotional response settings for their agents.

2.2.2 Multilingual & Real-Time Language Switching

ElevenLabs Conversational AI 2.0 includes automatic language detection allowing agents to identify the spoken language and respond appropriately within the same interaction. This capability should be exposed as a configuration option.

2.2.3 Agentic AI & Autonomous Workflows

Gartner predicts 40% of enterprise applications will integrate task-specific AI agents by end of 2026. Voice agents are evolving beyond Q&A to execute complex multi-step workflows autonomously. Our platform should support webhook-based actions and tool integrations.

2.2.4 Retrieval-Augmented Generation (RAG)

ElevenLabs has integrated RAG directly into the voice agent architecture, enabling retrieval from custom knowledge bases with minimal latency and maximum privacy. This is critical for domain-specific agents in healthcare, legal, and financial services.

2.2.5 Multimodal Capabilities

ElevenLabs Conversational AI 2.0 supports multimodality, allowing agents to communicate over text, voice, or both simultaneously. This creates a seamless omnichannel experience. Our platform should support both voice and text-based testing and interaction.

2.2.6 Voice Biometric Security

Voice biometric authentication provides frictionless security while reducing fraud. As a future enhancement, this feature would enable agent access control through voice verification.

2.2.7 Analytics & Conversation Intelligence

92% of businesses now capture speech data, with 56% transcribing more than half of their interactions. Comprehensive analytics dashboards tracking resolution rates, customer experience metrics, and compliance are becoming standard in production deployments.

Features Derived from Market Research (Recommended)

- Emotional tone configuration per agent (calm, empathetic, professional, energetic)
- Multilingual agent support with automatic language detection
- Knowledge base upload (PDF, CSV, TXT) with RAG-powered responses
- Webhook/API action triggers during conversations
- Conversation analytics dashboard with sentiment tracking
- Agent performance scoring and call quality metrics
- Voice cloning capability for brand-specific voices
- A/B testing between different agent configurations

3. Technical Architecture

3.1 Technology Stack

Layer	Technology	Purpose
Frontend Framework	Next.js 14+ (App Router)	Server-side rendering, routing, API routes
UI Library	React 18+	Component-based UI with hooks
UI Styling	Tailwind CSS + shadcn/ui	Rapid, consistent, accessible design system
State Management	Zustand / React Context	Client-side state management
Database	Supabase (PostgreSQL)	Relational data, real-time subscriptions
Authentication	Supabase Auth	Email/password, OAuth (Google, GitHub)
File Storage	Supabase Storage	Knowledge base documents, agent assets
Backend API	Next.js API Routes + Node.js	RESTful endpoints, server-side logic
AI Reasoning	Claude API (Anthropic)	Conversational intelligence, prompt engineering
Voice Synthesis	ElevenLabs API	Text-to-speech, voice agent, voice cloning
Real-time Comm	WebSockets (Supabase Realtime)	Live chat, agent testing, status updates
Deployment	Docker + Any Cloud (AWS/GCP/VPS)	Containerized, self-hosted deployment

3.2 System Architecture Diagram

The system follows a three-tier architecture with clear separation of concerns between the presentation layer (Next.js frontend), application layer (API routes and backend services), and data layer (Supabase and external APIs).

Component	Interactions	Protocol
Browser (React/Next.js)	API Routes, Supabase Auth, WebSocket	HTTPS, WSS
Next.js API Routes	Supabase DB, Claude API, ElevenLabs API	HTTPS, REST
Supabase	PostgreSQL, Auth, Storage, Realtime	Internal
Claude API	Receives prompts, returns reasoning	HTTPS REST
ElevenLabs API	Voice synthesis, agent management, RAG	HTTPS REST, WSS
Docker Container	Hosts Next.js app, Node.js backend	TCP/HTTP

3.3 API Keys & Environment Configuration

The following API keys and secrets are required for deployment. All keys must be stored as environment variables and never committed to version control.

Variable	Service	Required
NEXT_PUBLIC_SUPABASE_URL	Supabase Project URL	Yes
NEXT_PUBLIC_SUPABASE_ANON_KEY	Supabase Anonymous Key	Yes
SUPABASE_SERVICE_ROLE_KEY	Supabase Service Role Key	Yes
ANTHROPIC_API_KEY	Claude API Key	Yes
ELEVENLABS_API_KEY	ElevenLabs API Key	Yes
NEXTAUTH_SECRET	Session encryption secret	Yes
NEXT_PUBLIC_APP_URL	Application base URL	Yes

4. Detailed Feature Requirements

4.1 Module 1: Authentication & User Management

4.1.1 Sign Up Page

- Email and password registration via Supabase Auth
- OAuth integration (Google, GitHub) for one-click sign up
- Email verification workflow with magic link support
- Password strength validation (minimum 8 characters, mixed case, numbers)
- Terms of Service and Privacy Policy acceptance checkbox
- Redirect to onboarding wizard on first sign up

4.1.2 Sign In Page

- Email/password login with remember me option
- OAuth login (Google, GitHub)
- Forgot password flow with Supabase password reset email
- Session management with JWT refresh tokens via Supabase
- Rate limiting on failed login attempts (5 per minute)

4.1.3 User Profile

- Profile edit screen (name, avatar, company, timezone)

- API key management for programmatic access
- Account deletion with data purge option

4.2 Module 2: Landing Page

- Hero section with value proposition and CTA buttons
- Feature showcase section highlighting key platform capabilities
- Domain/industry use-case cards (Healthcare, Sales, Support, Education, etc.)
- Interactive demo section showing agent conversation preview
- Pricing section (Free Tier, Pro, Enterprise)
- Testimonials and social proof section
- Footer with links, legal, and contact information
- Fully responsive design for mobile, tablet, and desktop
- SEO-optimized with meta tags, Open Graph, and structured data

4.3 Module 3: Main Dashboard

4.3.1 Dashboard Overview

- Welcome card with user name and quick-start guide
- Agent summary cards showing total agents, active agents, total conversations
- Recent activity feed (agent created, tested, updated)
- Quick action buttons: Create New Agent, View Analytics, Manage Knowledge Base
- Usage metrics showing API calls consumed vs. plan limits

4.3.2 Agent Listing

- Grid/list toggle view of all created agents
- Each agent card shows: name, domain, status (draft/active/paused), voice type, last tested date
- Search and filter by domain, status, creation date
- Bulk actions: activate, pause, delete selected agents
- Pagination with 12 agents per page default

4.4 Module 4: AI Voice Agent Builder (Core Feature)

This is the primary feature of the platform. The agent builder follows a step-by-step wizard flow with the following configuration stages:

4.4.1 Step 1: Domain Selection

Users select the industry/domain their voice agent will operate in. This determines the default behavior templates and suggested configurations.

Domain	Description	Default Agent Types
Healthcare	Patient support, appointment scheduling, triage	Receptionist, Triage Assistant, Follow-Up Agent
Sales & Lead Gen	Outbound calls, lead qualification, follow-ups	SDR Agent, Lead Qualifier, Demo Scheduler
Customer Support	Inbound support, FAQ handling, ticket creation	Support Agent, FAQ Bot, Escalation Handler
Education	Student assistance, course info, enrollment	Enrollment Advisor, Tutor, Course Guide
Real Estate	Property inquiries, scheduling viewings	Property Agent, Viewing Scheduler, Buyer Qualifier
Hospitality	Reservations, guest services, concierge	Reservation Agent, Concierge, Feedback Collector
E-Commerce	Order tracking, returns, product recommendations	Order Support, Returns Agent, Product Advisor
Custom	User-defined domain with manual configuration	Custom Agent

4.4.2 Step 2: Agent Type & Persona Configuration

- **Agent Name:** Custom name for identification (e.g., "Dr. Sarah - Patient Intake")
- **Agent Type:** Select from domain-specific types or create custom
- **Agent Description:** Brief description of the agent’s purpose and capabilities
- **Personality Traits:** Select from predefined traits (Professional, Friendly, Empathetic, Authoritative, Casual)
- **Response Style:** Concise, Detailed, Conversational, Formal
- **First Message:** Customizable greeting message the agent uses to open conversations
- **System Prompt:** Advanced text area for precise behavioral instructions (with template suggestions)

4.4.3 Step 3: Voice Configuration

Leveraging ElevenLabs’ extensive voice library and synthesis capabilities:

- **Voice Gender:** Male, Female, or Neutral options

- **Voice Selection:** Browse and preview from ElevenLabs' 5,000+ voice library
- **Voice Preview:** Play sample audio before selecting
- **Language Selection:** Primary language with multilingual auto-detection option
- **Voice Cloning (Pro):** Upload audio samples to create custom brand voice
- **Speech Parameters:** Speed, stability, similarity boost, style exaggeration sliders
- **Accent/Dialect:** Regional accent options within selected language

4.4.4 Step 4: Knowledge Base & Training Data

This step configures the agent's domain knowledge using RAG (Retrieval-Augmented Generation):

- **Document Upload:** Support for PDF, DOCX, TXT, CSV, and JSON files (max 50MB per file)
- **URL Scraping:** Input website URLs to automatically extract and index content
- **FAQ Builder:** Manual question-answer pair entry interface
- **Knowledge Preview:** View indexed content with relevance scoring
- **Training Status:** Progress indicator showing indexing and embedding status
- **Custom Instructions:** Free-text area for specific behavioral rules and constraints

4.4.5 Step 5: Advanced Settings

- **LLM Model Selection:** Choose Claude model variant (Sonnet for speed, Opus for depth)
- **Temperature Setting:** 0.0–1.0 slider for response creativity vs. consistency
- **Max Conversation Length:** Set maximum turns or duration per conversation
- **Fallback Behavior:** Configure what happens when agent cannot answer (transfer, escalate, collect info)
- **Webhook URL:** POST endpoint for real-time conversation events
- **Allowed Topics:** Whitelist of topics the agent should discuss
- **Blocked Topics:** Blacklist of topics the agent must avoid
- **HIPAA Mode:** Enable for healthcare agents requiring compliance

4.4.6 Step 6: Review & Deploy

- Complete configuration summary across all steps
- Agent preview with simulated conversation
- Save as Draft or Deploy options
- Generate embeddable widget code for websites

- Generate shareable test link

4.5 Module 5: Agent Testing Interface

A floating chat widget that appears after agent deployment, enabling real-time testing:

4.5.1 Chat Widget UI

- Floating circular button in bottom-right corner with agent avatar
- Expandable chat window (400px wide, 600px tall) with smooth animation
- Agent name and status indicator (online/processing) in header
- Close, minimize, and fullscreen controls

4.5.2 Conversation Interface

- Real-time text transcript showing both user messages and agent responses
- User input: text field with send button AND microphone button for voice input
- Agent responses appear as text with optional audio playback button
- Typing/processing indicator with animated dots while agent generates response
- Timestamp on each message
- Message status indicators (sent, processing, delivered, error)

4.5.3 Voice Interaction

- Push-to-talk or continuous listening modes
- Real-time speech-to-text display as user speaks
- Agent response auto-plays as audio via ElevenLabs TTS
- Volume control and mute option
- Visual audio waveform during playback

4.5.4 Testing Features

- Conversation reset button to start fresh session
- Export conversation as JSON or PDF
- Rate individual responses (thumbs up/down) for quality tracking
- Debug panel showing: prompt sent to Claude, tokens used, response latency, voice generation time
- Share test session link with team members

5. Database Schema (Supabase/PostgreSQL)

5.1 Core Tables

`users` — Extended user profile (supplements Supabase auth.users)

Column	Type	Description
id	UUID (PK, FK → auth.users)	Primary key linked to Supabase auth
full_name	VARCHAR(255)	User display name
avatar_url	TEXT	Profile image URL
company	VARCHAR(255)	Company/organization name
timezone	VARCHAR(100)	User's timezone preference
plan_tier	ENUM(free, pro, enterprise)	Subscription tier
api_calls_used	INTEGER	Monthly API call counter
api_calls_limit	INTEGER	Monthly API call limit
created_at	TIMESTAMPTZ	Account creation date
updated_at	TIMESTAMPTZ	Last profile update

`agents` — AI voice agent configurations

Column	Type	Description
id	UUID (PK)	Unique agent identifier
user_id	UUID (FK → users)	Owner of the agent
name	VARCHAR(255)	Agent display name
description	TEXT	Agent purpose description
domain	VARCHAR(100)	Business domain/industry
agent_type	VARCHAR(100)	Type within domain
status	ENUM(draft, active, paused, archived)	Current agent status
personality_traits	JSONB	Personality configuration object
system_prompt	TEXT	System prompt for Claude API
first_message	TEXT	Greeting message
voice_id	VARCHAR(255)	ElevenLabs voice identifier
voice_gender	VARCHAR(20)	Male/Female/Neutral
language	VARCHAR(50)	Primary language code
voice_settings	JSONB	Speed, stability, style params

Column	Type	Description
llm_model	VARCHAR(100)	Claude model variant
temperature	DECIMAL(3,2)	LLM temperature setting
max_turns	INTEGER	Max conversation turns
webhook_url	TEXT	Event notification endpoint
advanced_settings	JSONB	Allowed/blocked topics, etc.
elevenlabs_agent_id	VARCHAR(255)	Mapped ElevenLabs agent ID
created_at	TIMESTAMPTZ	Creation timestamp
updated_at	TIMESTAMPTZ	Last update timestamp

`knowledge_bases` — Training data and documents per agent

Column	Type	Description
id	UUID (PK)	Knowledge item identifier
agent_id	UUID (FK → agents)	Linked agent
source_type	ENUM(file, url, faq, text)	Type of knowledge source
file_path	TEXT	Supabase storage path (if file)
source_url	TEXT	Original URL (if scraped)
content	TEXT	Extracted/raw content
metadata	JSONB	File size, page count, etc.
indexing_status	ENUM(pending, processing, indexed, failed)	Processing status
created_at	TIMESTAMPTZ	Upload timestamp

`conversations` — Testing and production conversation sessions

Column	Type	Description
id	UUID (PK)	Conversation session ID
agent_id	UUID (FK → agents)	Agent involved
user_id	UUID (FK → users)	User who initiated
status	ENUM(active, completed, error)	Session status
total_turns	INTEGER	Number of exchanges
total_tokens	INTEGER	Total tokens consumed
duration_seconds	INTEGER	Total conversation duration
sentiment_score	DECIMAL(3,2)	Overall sentiment (-1 to 1)

Column	Type	Description
started_at	TIMESTAMPTZ	Session start time
ended_at	TIMESTAMPTZ	Session end time

`messages` — Individual messages within conversations

Column	Type	Description
id	UUID (PK)	Message identifier
conversation_id	UUID (FK → conversations)	Parent conversation
role	ENUM(user, agent, system)	Message sender role
content	TEXT	Message text content
audio_url	TEXT	ElevenLabs audio file URL
tokens_used	INTEGER	Tokens for this message
latency_ms	INTEGER	Response generation time
rating	SMALLINT	User rating (1 = bad, 5 = good)
created_at	TIMESTAMPTZ	Message timestamp

5.2 Row Level Security (RLS)

All tables must implement Supabase Row Level Security policies to ensure multi-tenant data isolation. Every query is automatically scoped to the authenticated user's ID, preventing cross-user data access.

- Users can only read/write their own profile data
- Agents are accessible only to their creator (user_id match)
- Knowledge bases inherit agent-level access control
- Conversations and messages are scoped to the agent owner
- Service role key bypasses RLS for server-side operations only

6. API Endpoint Specification

6.1 Authentication Endpoints

Method	Endpoint	Description
POST	/api/auth/signup	Register new user

Method	Endpoint	Description
POST	/api/auth/signin	Authenticate existing user
POST	/api/auth/signout	Invalidate session
POST	/api/auth/reset-password	Send password reset email
GET	/api/auth/me	Get current user profile
PATCH	/api/auth/profile	Update user profile

6.2 Agent Management Endpoints

Method	Endpoint	Description
GET	/api/agents	List all agents for authenticated user
POST	/api/agents	Create new agent with configuration
GET	/api/agents/:id	Get single agent details
PATCH	/api/agents/:id	Update agent configuration
DELETE	/api/agents/:id	Delete agent and associated data
POST	/api/agents/:id/deploy	Deploy/activate agent
POST	/api/agents/:id/pause	Pause active agent
POST	/api/agents/:id/duplicate	Clone agent configuration

6.3 Knowledge Base Endpoints

Method	Endpoint	Description
GET	/api/agents/:id/knowledge	List knowledge items for agent
POST	/api/agents/:id/knowledge/upload	Upload document file
POST	/api/agents/:id/knowledge/url	Add URL for scraping
POST	/api/agents/:id/knowledge/faq	Add FAQ question-answer pair
DELETE	/api/agents/:id/knowledge/:kid	Remove knowledge item
GET	/api/agents/:id/knowledge/status	Check indexing status

6.4 Conversation & Testing Endpoints

Method	Endpoint	Description
POST	/api/agents/:id/chat	Send message, get agent response
POST	/api/agents/:id/chat/voice	Send audio, get voice response

Method	Endpoint	Description
GET	/api/agents/:id/conversations	List conversation history
GET	/api/conversations/:cid	Get full conversation transcript
POST	/api/conversations/:cid/rate	Rate a message in conversation
DELETE	/api/conversations/:cid	Delete conversation session

6.5 Analytics Endpoints

Method	Endpoint	Description
GET	/api/analytics/overview	Dashboard summary metrics
GET	/api/analytics/agents/:id	Per-agent performance metrics
GET	/api/analytics/usage	API usage and billing data

6.6 ElevenLabs Integration Endpoints

Method	Endpoint	Description
GET	/api/voices	List available ElevenLabs voices
GET	/api/voices/:id/preview	Get voice preview audio
POST	/api/voices/clone	Upload audio for voice cloning
GET	/api/voices/languages	List supported languages

7. UI/UX Wireframe Specification

7.1 Page Map

Page	Route	Auth Required	Purpose
Landing Page	/	No	Marketing, features, pricing, CTA
Sign Up	/signup	No	New user registration
Sign In	/signin	No	User authentication
Dashboard	/dashboard	Yes	Agent overview, quick actions

Page	Route	Auth Required	Purpose
Agent Builder	/agents/new	Yes	Step-by-step agent creation wizard
Agent Detail	/agents/:id	Yes	View/edit agent configuration
Agent Testing	/agents/:id/test	Yes	Interactive chat testing interface
Knowledge Base	/agents/:id/knowledge	Yes	Manage training data
Analytics	/analytics	Yes	Usage and performance dashboards
Profile	/profile	Yes	User settings and account management
Billing	/billing	Yes	Plan management and usage

7.2 Design System Guidelines

- Color Palette: Primary Navy (#1B2A4A), Accent Blue (#2E75B6), Success Green (#27AE60), Warning Amber (#F39C12), Error Red (#E74C3C)
- Typography: Inter or DM Sans for UI, JetBrains Mono for code/technical elements
- Spacing: 4px base grid system (4, 8, 12, 16, 24, 32, 48, 64)
- Border Radius: 8px for cards, 6px for buttons, 4px for inputs
- Shadows: Subtle elevation system (sm, md, lg) for depth hierarchy
- Animations: Smooth 200ms transitions for interactions, 300ms for modals
- Accessibility: WCAG 2.1 AA compliance, keyboard navigation, screen reader support
- Dark Mode: Full dark theme support with automatic system preference detection

8. ElevenLabs & Claude API Integration Flow

8.1 Agent Creation Flow

1. User completes agent configuration wizard in the frontend
2. Backend saves configuration to Supabase agents table
3. Backend calls ElevenLabs API to create a Conversational AI agent with the configured voice, language, and system prompt
4. If knowledge base documents exist, backend uploads them to ElevenLabs RAG or processes them via Claude for embedding
5. ElevenLabs returns an agent_id which is stored in the agents table

6. Agent status updated to “active” and testing interface becomes available

8.2 Conversation Flow (Testing)

1. User opens chat widget and speaks or types a message
2. If voice input: browser captures audio via Web Audio API and sends to backend
3. Backend transcribes audio using ElevenLabs Speech-to-Text (or sends text directly)
4. Transcribed text + conversation history + system prompt sent to Claude API
5. Claude generates contextual response based on agent configuration and knowledge base
6. Response text sent to ElevenLabs TTS API with configured voice_id and settings
7. Audio stream returned to frontend; text transcript displayed simultaneously
8. Conversation turn saved to messages table with latency metrics

9. Non-Functional Requirements

Category	Requirement	Target
Performance	Page load time (LCP)	< 2.5 seconds
Performance	API response time (p95)	< 500ms (excluding AI calls)
Performance	Voice agent response latency	< 2 seconds end-to-end
Scalability	Concurrent users supported	1,000+ simultaneous
Scalability	Agents per user	Up to 50 (Pro), unlimited (Enterprise)
Availability	Uptime SLA target	99.5%
Security	Data encryption	AES-256 at rest, TLS 1.3 in transit
Security	Authentication tokens	JWT with 1-hour expiry, refresh rotation
Security	API rate limiting	100 req/min (Free), 500 (Pro), custom (Enterprise)
Compliance	Data privacy	GDPR-ready, data residency options
Compliance	Healthcare agents	HIPAA-compliant mode via ElevenLabs
Accessibility	WCAG compliance	Level AA minimum
Browser Support	Minimum versions	Chrome 90+, Firefox 90+, Safari 15+, Edge 90+

10. Deployment & Infrastructure

10.1 Docker Deployment

The application will be containerized using Docker for consistent deployment across any infrastructure. A docker-compose.yml file will orchestrate the Next.js application alongside any required services.

- Multi-stage Dockerfile for optimized production builds
- Docker Compose for local development and single-server deployment
- Environment variable injection via .env files or secrets management
- Health check endpoints for container orchestration
- Automatic restart policies for production stability

10.2 Recommended Hosting Options

Provider	Service	Best For	Est. Monthly Cost
AWS	EC2 + RDS / ECS Fargate	Enterprise, auto-scaling	$50–$500+
Google Cloud	Cloud Run / GKE	Serverless, Kubernetes	$30–$300+
DigitalOcean	App Platform / Droplets	Simple, cost-effective	$20–$100
Railway	Managed Deployment	Developer-friendly, fast deploy	$10–$50
Vercel + Supabase	Edge + Managed DB	Next.js optimized, rapid iteration	$20–$100
Self-hosted VPS	Any Linux VPS	Full control, data sovereignty	$10–$50

11. Project Timeline & Milestones

Phase	Duration	Deliverables
Phase 1: Foundation	Week 1–2	Project setup, Supabase config, auth flow, landing page, basic UI shell
Phase 2: Dashboard	Week 3–4	Dashboard layout, agent listing, user profile, navigation
Phase 3: Agent Builder	Week 5–7	6-step wizard, ElevenLabs voice integration, knowledge base upload
Phase 4: Chat & Testing	Week 8–9	Chat widget, voice input/output, conversation transcripts, real-time testing
Phase 5: Analytics	Week 10	Analytics dashboard, usage tracking, conversation metrics

Phase	Duration	Deliverables
Phase 6: Polish & Deploy	Week 11–12	UI polish, Docker setup, deployment scripts, documentation, testing

Total Estimated Timeline: 12 weeks (3 months) for MVP with core features. Additional 4–6 weeks for advanced features (voice cloning, A/B testing, webhook integrations, white-label options).

12. Suggested Pricing Model

Feature	Free Tier	Pro ($49/mo)	Enterprise (Custom)
Voice Agents	2	25	Unlimited
Conversations/Month	100	5,000	Unlimited
Knowledge Base Size	5 MB	500 MB	Unlimited
Voice Cloning	No	Yes (5 voices)	Yes (unlimited)
Voice Library Access	Basic (50 voices)	Full (5,000+)	Full + Custom
Analytics	Basic	Advanced	Advanced + Export
Webhook Integrations	No	Yes	Yes + Priority
Support	Community	Email	Dedicated Account Manager
API Access	No	Yes	Yes + SLA
White-Label Branding	No	No	Yes
HIPAA Compliance	No	Add-on ($29/mo)	Included

13. Risks & Mitigations

Risk	Impact	Likelihood	Mitigation
ElevenLabs API rate limits	High	Medium	Implement queue system, caching, and fallback voices
Claude API cost overrun	High	Medium	Token budgeting per agent, usage alerts, model tier selection
Voice latency > 3 seconds	Medium	Medium	Use ElevenLabs Turbo models, stream audio chunks, edge caching

Risk	Impact	Likelihood	Mitigation
Multi-tenant data leak	Critical	Low	Supabase RLS policies, penetration testing, audit logging
Third-party API downtime	High	Low	Health checks, graceful degradation, status page monitoring
Knowledge base indexing failures	Medium	Medium	Retry logic, file validation, user notification on failure
Browser mic permission issues	Low	High	Clear permission prompts, text-only fallback, browser compatibility guide

14. Future Enhancements (Post-MVP Roadmap)

14.1 Phase 2 Features (Months 4–6)

- Telephony Integration: Connect agents to phone numbers via Twilio/ElevenLabs SIP trunking for inbound/outbound calls
- WhatsApp & SMS Channels: Deploy agents on messaging platforms for omnichannel reach
- A/B Testing Framework: Compare agent configurations and measure performance differences
- Team Collaboration: Multi-user workspaces with role-based access control (admin, editor, viewer)
- Conversation Handoff: Seamless transfer from AI agent to human agent with full context

14.2 Phase 3 Features (Months 7–12)

- White-Label Solution: Complete branding customization for resellers and agencies
- Marketplace: Community-shared agent templates, voice packs, and knowledge base modules
- Advanced Analytics: Sentiment analysis trends, conversion funnels, customer satisfaction scoring
- Multi-Agent Workflows: Chain multiple agents for complex conversation flows
- Voice Biometric Authentication: Speaker verification for secure agent interactions
- CRM Integrations: Native connectors for Salesforce, HubSpot, Zendesk, and Freshdesk
- Mobile SDK: Embeddable voice agent for iOS and Android applications

15. Appendix

15.1 Glossary

Term	Definition
TTS	Text-to-Speech: Converting written text to spoken audio
STT	Speech-to-Text: Converting spoken audio to written text
RAG	Retrieval-Augmented Generation: Enhancing AI responses with external knowledge
LLM	Large Language Model: AI model for understanding and generating text (e.g., Claude)
RLS	Row Level Security: Database-level access control policy (Supabase)
VAD	Voice Activity Detection: Detecting when a user starts/stops speaking
NLP	Natural Language Processing: AI understanding of human language
CAGR	Compound Annual Growth Rate: Yearly growth metric
SIP	Session Initiation Protocol: Standard for voice/video calling over internet
JWT	JSON Web Token: Secure authentication token format

15.2 Reference Links

- ElevenLabs Documentation: https://elevenlabs.io/docs
- ElevenLabs Conversational AI: https://elevenlabs.io/conversational-ai
- Claude API Documentation: https://docs.anthropic.com
- Supabase Documentation: https://supabase.com/docs
- Next.js Documentation: https://nextjs.org/docs
- Tailwind CSS: https://tailwindcss.com/docs
- shadcn/ui Components: https://ui.shadcn.com

End of Document

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